



UNIVERSIDAD TECNICA FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

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Enero 2025 , Valparaíso, Chile



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Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This paper presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este artículo se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

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Chapter 1

Introduction

1.1 Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5% of energy sector CO₂ emissions [1] [2], has continuously increased in emissions, as shown in figure 1.1.

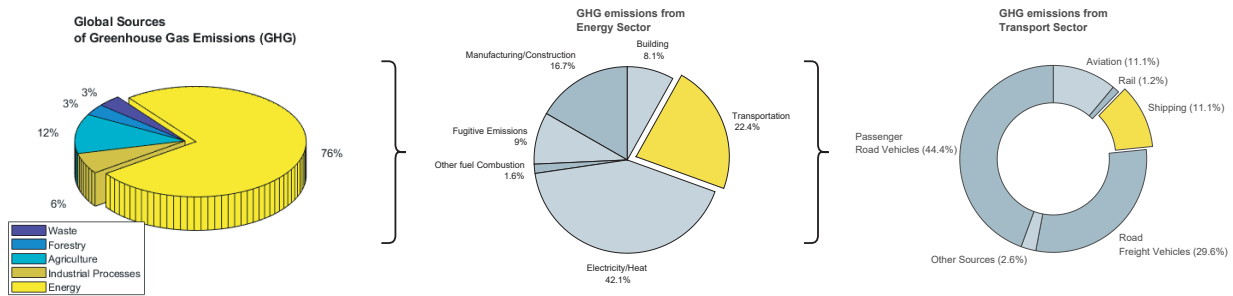


Figure 1.1: Global sources of greenhouse gas emissions and breakdown of CO₂ emissions from the energy sector and the transport subsector [1].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and providing incentives. The initial strategy avoids a reduction in total GHG emissions from international shipping by at least 50% by 2030 compared to 2008, consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5% mass by mass (m/m) by 2025 per the International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI, as shown in figure 1.2a. Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed, as shown in figure 1.2b. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min.

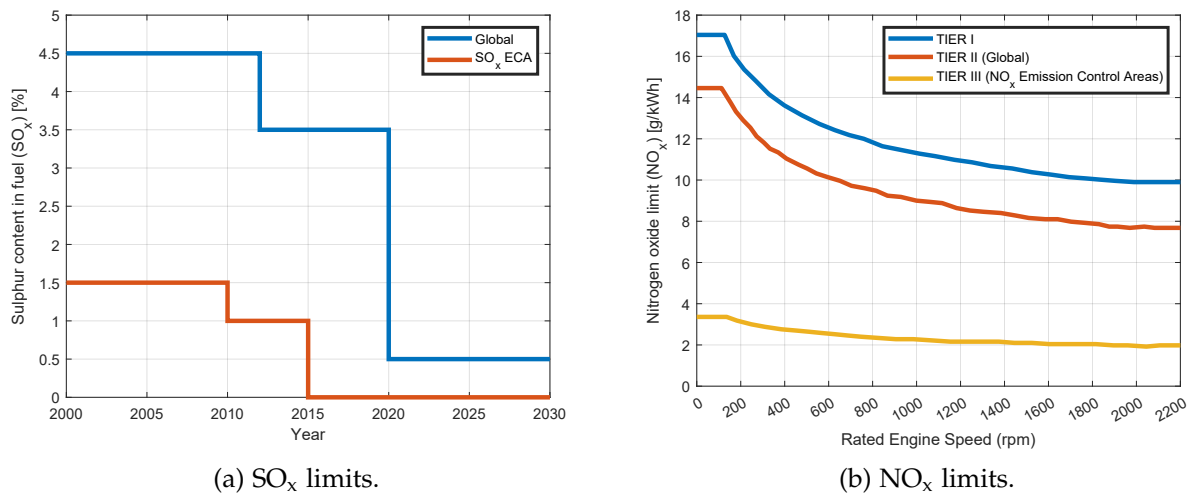


Figure 1.2: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs perspective).

To comply with the standards and emissions reduction goals set by IMO, the maritime industry is looking for reduced and zero-emission technologies and approaches. The transition from traditional fossil fuel engines to zero-emissions technologies is challenging in a competitive market, where manufacturers are struggling to increase revenue and reduce production cost. The incentives to reduce emissions and improve the efficiency of equipment are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storages (for example, batteries and supercapacitors) reduce GHG emissions and are the key enablers for the realization of low-emissions operation of marine vessels.

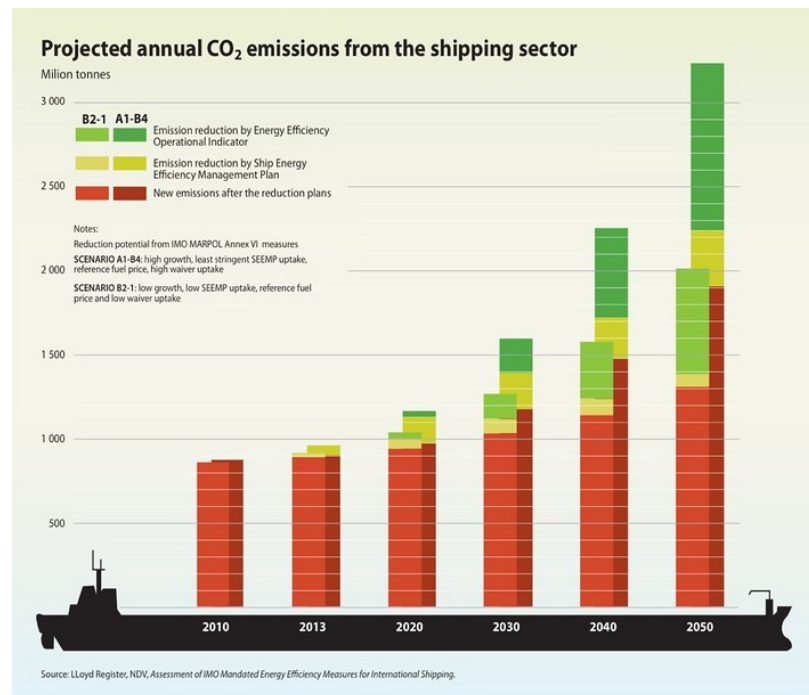


Figure 1.3: IMO greenhouse gas emissions strategy [6].

1.1.1 Propulsion Architectures

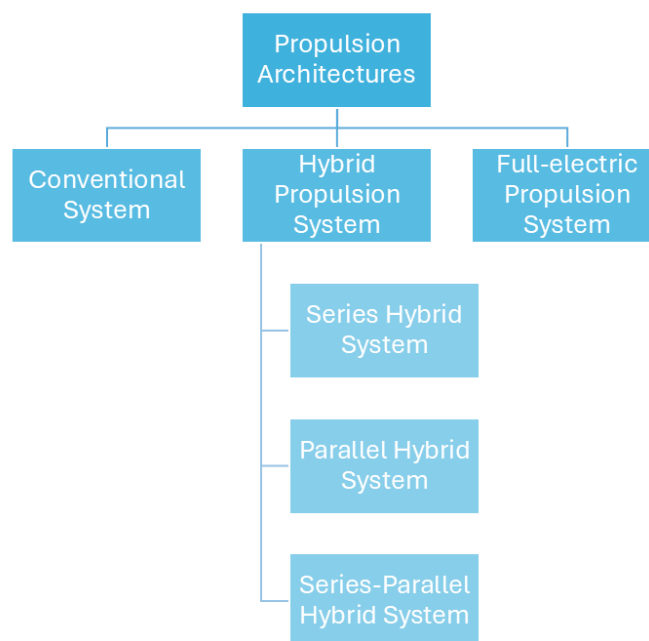


Figure 1.4: Types of propulsion systems-- > PDF

Full-electric and hybrid propulsion architectures, allowed by innovative and high performance energy storage (batteries, fuel cells, supercapacitors, etc) are a promising track to reduce strongly pollutants and GHG emissions [4]. New developments of electrically propelled ships are based on an integrated energy distribution system architecture that releases large propulsion machines [5].

For full-electric propulsion, the main advantages include better dynamic performance (start-up, arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because only the electric motor must be installed close to the propeller.

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds [5]. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. Hybrid propulsion systems (HPS) are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significantly for different types of vessel, mainly due to the different types of mission profiles and operation requirements [4].

Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources. The choice of different sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

As can be seen in the following subsections, there are three main architectures for hybrid propulsion chains [4]:

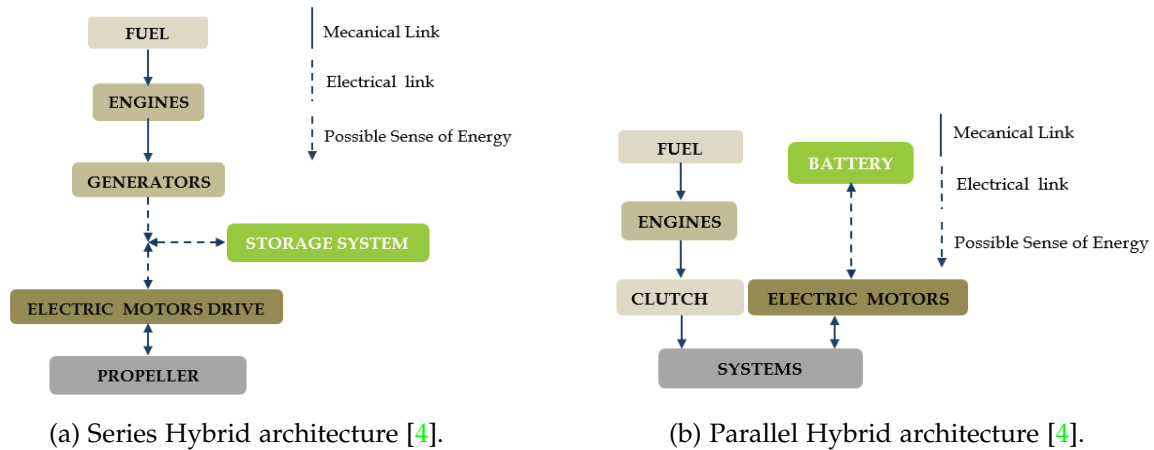


Figure 1.5: Semi-hybridization.

a) Series Hybrid Architectures

In this type of HPS architecture, two kind of energy system (combustion engines and electric systems) are used in series. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter), as can be seen in the figure 1.5a. The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines. Another option is to use fuel cells to provide electric power or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators. In this configuration, electric power is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

b) Parallel Hybrid Architectures

Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches, as can be seen in the figure 1.5b. The electrical motors and drive are supplied by electrical ESS and/or independent power sources generators driven by combustion engines and/or Fuel Cell for example. The two propulsion systems can be used together or separately. In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources.

c) Series-Parallel Hybrid Architectures

The series-parallel architecture combines both architectures defined previously. This association allows the passage of a parallel-type operation to a serial type operation.

Hybrid Propulsion systems are based on the combination of several power sources and energy storage systems.

1.1.2 Power Generation Systems

a) Steam Turbine

b) Gas Turbine

c) Diesel Engines

A diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [4]. Diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [9]. Compared to other prime movers such as a gas turbine or steam turbine, the diesel engine has a high efficiency. Its low fuel consumption is the main reason it is widely used in marine applications.

This power source is characterized by a high level of autonomy and by a relatively low cost compared to other sources. Diesel engines are characterized by a high level of greenhouse gases and pollutants emission particularly if they are used at a lower level of power than their rated power.

1.1.3 Energy Storage Systems

a) Supercapacitors

Supercapacitors can be interesting intermediate energy storage systems for small ships with short power transients [4]. They are characterized by a high value of charge and discharge peak power and can be used to rapid dynamic storage with very low specific energy. They are usually associated with slower systems, such as batteries and fuel cells, to manage high power short transients. They can operate at very low temperatures at which many types of electrochemical batteries stop working.

b) Flywheels

Flywheels can be used in some particular cases. Like supercapacitors, they are characterized by a high power density but a relatively low energy density. However using flywheels leads to implement safety systems to protect crew and passengers from possible flywheel failure. So their use in small ship context seems very limited.

c) Batteries

The oldest and most classical method for storing electrical energy is using batteries. Battery technology has matured significantly during the past two decades as not only the world but also energy has become increasingly mobile [16].

Table 1.1: Energy Storage Systems Comparison [17]

Technology	Battery	Supercapacitor	Flywheel
Energy Density (Wh/kg)	50-250	0.05-5	5-100
Power Density (W/Kg)	50-2000	100k	1000
Lifecycle (cycle)	400-9000	50k	20k-100k
Lifetime (years)	5-10	5-8	15-20
Overall Efficiency (%)	85-99	60-65	15-20

1.1.4 Battery Systems in Maritime Applications

In 2018, the European Maritime Safety Agency (EMSA) commissioned DNV GL to perform a study on the use of electrical storage systems in shipping, with the objective of providing an overview of technology, research, feasibility, regulations and safety of battery systems in maritime applications [15].

Battery application onboard ships can have multiple functional roles. While batteries can fully power a vessel for short distance or duration, improving performance and energy efficiency of the overall vessel is often the key purpose. Among the different uses of batteries on ships are:

- Spinning reserve
 - Backup for running generators.
 - Fewer generators needed online.
- Peak shaving
 - Act as a buffer.
 - Level power seen by engines.
- Optimise load
 - Optimise the operating point of the generators.
 - Reduce maintenance.
- Immediate power
 - Instant power in support of generators.
- Harvest energy
 - Recover energy from cranes, drilling equipment, etc.
 - Accommodate energy from renewables.
- Backup power
 - Battery system provides backup power, UPS like functionality.

Table 1.2: Energy Storage Systems Comparison [17]

Battery	Energy Density (Wh/kg)	Vol. Energy Density (Wh/L)	Cost (USD/kWh)	Lifecycle	Efficiency (%)
Lead-Acid	20-40	60-110	165	500-1000	50-95
Ni-Cd	20-60	60-150	-	2000-2500	70-90
Ni-MH	50-70	140-300	146	500-2000	89-92
Na-S	~120	150-300	400	>1500	86-92
Li-ion	50-250	250-675	200-600	400-9000	85-99
Li-polymer	~150	250-730	-	-	-
Li-S	400	350	400-500	1500	-

Different ships have different operating profiles and batteries must respond to specific energy and power demand while also having in consideration the desired/expected life-cycle for the battery [15].

Considering the incorporation of batteries into the propulsion system of vessels, different propulsion architectures are recognized that provide a different use for batteries, even without considering them, as is the case with conventional propulsion architecture [15] [16]. This can be seen in the figures 1.6, 1.7 and 1.8.

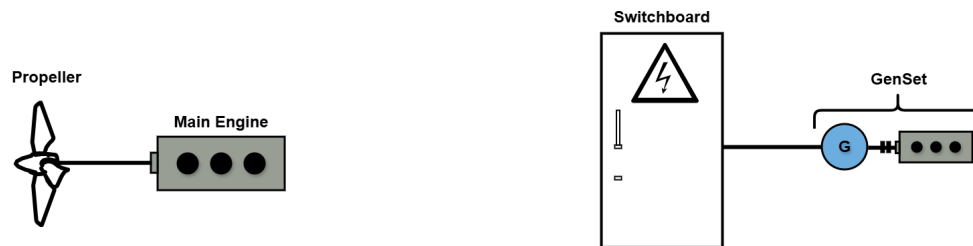
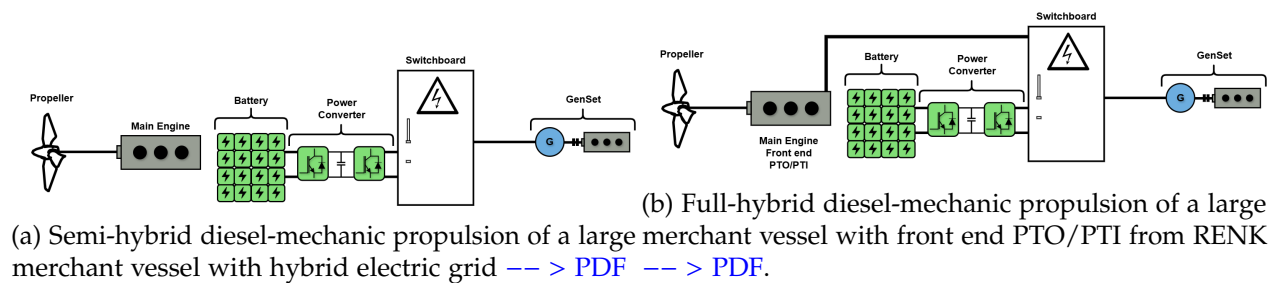


Figure 1.6: Traditional diesel-mechanic propulsion of a large merchant vessel -- > PDF.



(a) Semi-hybrid diesel-mechanic propulsion of a large merchant vessel with hybrid electric grid -- > PDF -- > PDF.

Figure 1.7: Semi-hybridization.

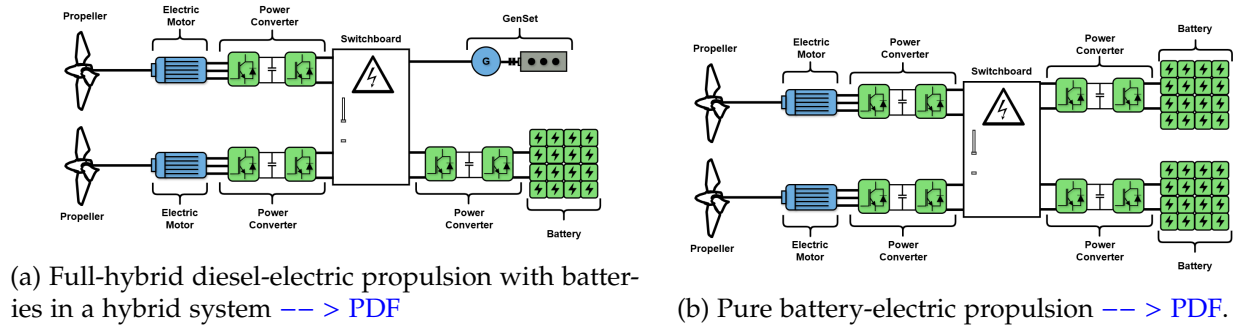


Figure 1.8: Electric motor incorporation.

1.1.5 Tugboats

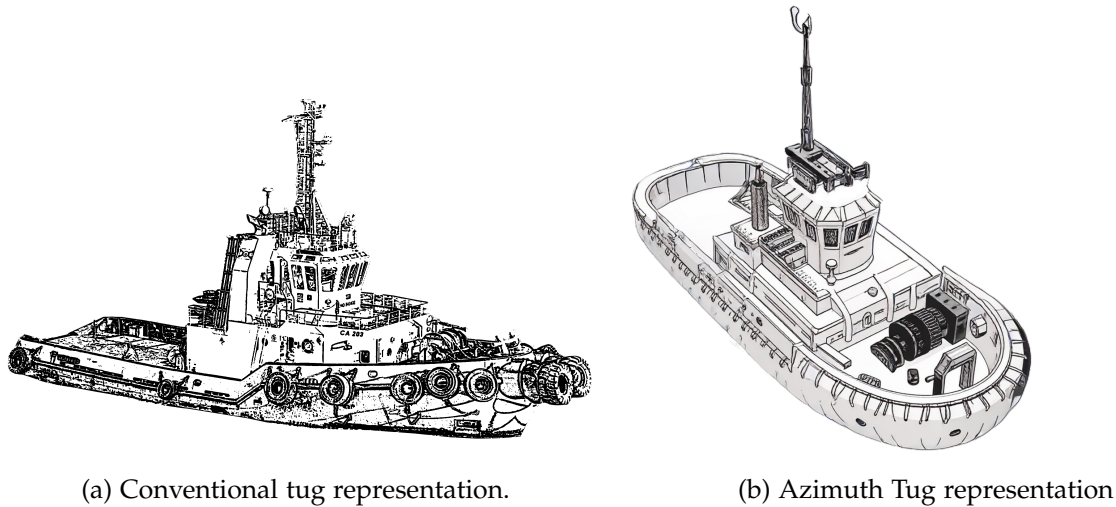


Figure 1.9: Tugs representations

Tugboats are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [17]. Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [3].

A tug, or more commonly a tugboat, is an assistance boat which helps in the mooring or berthing operation of a ship by either towing or pushing a vessel towards the port [10] [11].

A tug is a particular class of boat which helps mega-ships enter or leave a port. In addition to towing the vessel towards the harbour, tug boats can also be engaged to provide essentials, such as water, air, etc., to a vessel.

Tug boats are also essential for non-self-propelled barges, oil platforms, log rafts, etc. Due to their solid structural engineering, tugs are small but relatively powerful machines.

The usage and functions of tugs vary from port to port, as different ports have different requirements and intakes. The common thing is pushing or towing mega boats or barges. Their usage depends on the following factors:

- Port traffic Volume

- Types of ships to be served by that tug
- Navigational obstacles to be catered to
- Conditions of environmental protection
- Local laws and
- Domains to be carried by a tug

An average tug boat has 500-2500 kW engines (680-3400 hp), but those which are larger and venture out into deep waters have engines with a power close to 20000 kW (27200 hp) and a power/tonnage ratio ranging between 2.20-4.50 for large tugs and 4.0-9.5 for harbour tugs. These are high ratios, considering the ratio of cargo ships or vessels varies between 0.35 and 1.20.

Based on purpose, they serve marine tugs can be of two types:

- Escort Tugs: The tugs designed generally to escort and manoeuvre ferries and barges to their destination are known as escort tugs.
- Support Tugs: These tugs provide support services for offshore and towing operations. These tugs play a significant role in berthing operations.

Mainly, tugs used in the marine industry are of three types, which are listed:

1. Conventional tug
2. Tractor tug
3. Azimuth tug

The efficiency of these types of vessels is based on fuel consumption in relation to load capacities, always taking into account both the propulsion system and the energy conversion system that determines it. The efficiency of conventional propulsion systems varies between 38% and 41% for a power range between 100 kW and 2,500 kW. For vessels with higher power ratings, efficiency increases to 48% [12] [13]. Low efficiency is further affected by the variability of the operating and load conditions to which engines are subjected. This variability leads to an increase in specific fuel consumption, which causes higher emissions and energy waste through exhaust gases and cooling systems [12]. However, this level of efficiency could be optimized by the hybridization.

1.2 Challenges and Research Opportunities

Several studies have examined hybrid propulsion systems [8]. A genetic algorithm to obtain the optimal operating points regard with series hybrid configurations in power generation, especially for the case of heavy duty hybrid drive applications [14]. A marine hybrid propulsion system, focusing on vector control of the electric motor during different modes and verifying the control feasibility [18]. The optimization of hybrid propulsion system design for a tugboat has been explored, presenting a methodology that streamlines powertrain component sizing and control, minimizing costs for a specific operating profile [19]. A coordinated control strategy for a variable-speed hybrid tugboat have been presented to improve fuel economy. The proposed strategy, validated through simulations and a smallscale experimental testbed, showed reduced costs and

lower CO₂ emissions [20].

However, there is no clear and detailed methodology for the hybridization of vessels such as tugboats that would provide a path from the conceptual stage to the technical details and subsequent implementation.

1.3 Thesis Objectives and Outline

The general objective of this work is to develop a methodology for the design and sizing of the battery bank for a hybrid tugboat, which will allow us to determine the benefits that would be obtained from hybridization.

Specific objectives include:

- Model the operation of a conventional tugboat
- Define hybrid architecture.
- Propose the service power of the electrical system.
- Preliminary design of the battery bank based on technical and economic criteria.
- Simulation of the proposed cell and battery bank.

1.4 Methodology

The methodology to follow for the development of this work is:

- **State of Art:** Review of existing hybrid propulsion system types and battery technologies.
- **Mathematical Analysis and Modeling:** Theoretical study of the tugboat's load profile and the implementation of batteries to support the vessel's operation.
- **Results:** Simulation of the models obtained to complete the conceptual analysis.

1.5 Scope and Limitations

1.6 Chapter Summary

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